Tristen Chen

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EDUCATION & AWARDS

University of Toronto - *B.A.Sc. in Mechanical Engineering + PEY Co-op* Sept 2024-June 2029

- **Awards & Certificates:** Dean's Honour List, Wallace Chalmers Engineering Design Award, 1st in Battle of the School Hackathon simulation track, VEX Robotics World Championship qualifier & Design Award, SolidWorks CSWA, Basic Machining Certificate

SKILLS

- **CAD & Engineering Analysis:** Fusion 360, Inventor, AutoCAD, **SolidWorks (CSWA, FEA, Drawings, Assembly management)**, GD&T, **First-principles Failure Analysis**, Gearbox & Housing Design, **BoM & Component Sourcing**
- **Mechatronics:** ESP32, **Servo/Stepper Motors Control**, Circuit Prototyping, Sensor integration (UART)
- **Manufacturing & Prototyping**: Machining (Turning/ Milling), **FDM 3D printing**, Soldering
- **Programming & Simulation:** C/C++, MATLAB, **Python (OpenCV, MediaPipe, NumPy, SciPy),** Rerun

PROJECTS

Project Founder - *Stroke Bot: High-speed CoreXY Pen Plotter* Sept 2026-Present

- Won 1st place out of 15 teams at a 24-hr hackathon with an end-to-end vision-to-drawing motion simulation. Conceptualized and expanded the project into an ultra-fast CoreXY CNC Drawing machine.
- Designing a rigid CoreXY gantry in SolidWorks for 800 mm/s speed and 20,000 mm/s^2 acceleration; sizing stepper torque, belt routing, and side-mounted rails, and customizing mount from first principles to minimize moving mass and deflection.
- Built a Python pipeline (OpenCV, SciPy, NumPy) that converts image contours into dynamic joint poses via a warm-started inverse kinematics solver, validating spatial trajectories in Rerun before hardware execution.

Project Founder - *Trace Project: Soft Robotics Telepresence System* April 2026-Present

- Identified a gap in remote physical interaction and founded a 3-person multidisciplinary team (Mechanical, Electrical, Software), driving 4 rapid hardware iterations to showcase a working tendon-driven telepresence robot at Socratica.
- Diagnosed lateral-torsional buckling in an initial 3D-printed leaf spring through physical testing, redesigning the compliant joint with an anisotropic FR4 fiberglass spine. Validated the new design against stall torque and fatigue failure using SolidWorks FEA and first-principles-based hands-on calculations.
- Engineered a compact 6:1 planetary gearbox (1.8Nm holding torque, 150+ downloads on GrabCAD) and a double-layer snap-fit motor stack with Bowden cable routing, cutting horizontal footprint by 30% and eliminating tendon-path variation. Designed through SolidWorks and the Fusion 360 Gear generator.
- Developed a multithreaded Python (MediaPipe) hand-tracking pipeline over UDP with Kalman filtering, reducing end-to-end latency to sub-200ms.

Project Co-Lead - *MIE243 Cinematography Robotic Arm Design Project* Sept 2025-Dec 2025

- Co-led mechanical architecture and identified load cases and ISO safety factors; owned the motorized mounting interface from concept to production, delivering a 20+ part CAD assembly, full GD&T manufacturing drawings, and a complete BOM.

EXPERIENCE

Senior Optomechanical Engineer - *Payload Subteam, UTAT Space System, University of Toronto* Jan 2026–Present

- Engineered an internal pressure venting system with a 3x safety factor for a CubeSat optical housing. Modeled hole-to-volume ratios per NASA and NANOCUB standards, evaluated manufacturing methods, and produced technical reports documenting the process to prevent lens fracture by launch pressure differential.
- Directed SOLIDWORKS version control for a 20-person engineering team using Sibe PDM's check-out system, repairing broken assembly mates, and ensuring all modifications were integrated without redundancy.

Team Lead - *EWB Smart Watering System, University of Toronto* Oct 2025-April 2026

- Translated stakeholder needs into requirements and led a 4-person cross-disciplinary team across the full product lifecycle. Programmed C/C++ control loops for an ESP32 and integrated pumps, sensors, and housing to automatically collect and process weather and environmental data to fully automate the watering process for the Hart House Community Garden.

Automotive Manufacturing Intern - *Taiwan Isuzu Motors Co., Ltd* May 2025-Aug 2025

- Participated in the assembly of 10+ commercial trucks daily within a 20+ person SKD production line. Executed rigorous quality control inspections, interpreted complex assembly drawings to prevent defects and ensure strict compliance with Isuzu manufacturing standards.

Fabrication Team Member– *Chem-E-Cube, University of Toronto(Final qualifier, 13th/60+ teams)* Dec 2024–Oct 2025

- Independently owned the cooling architecture for a 1ft^3 Direct Air Capture Cube, optimizing the Monolith's CO2 absorption rate through a low-current Peltier control strategy integrated with a custom 3D-printed air channel to achieve 160% cooling efficiency.
- Designed airtight CO2 and temperature sensor housings in Inventor to obtain real-time data during operation. Utilized design for additive manufacturing principles to eliminate print supports and integrated locator pins for epoxy joining.